

COMP 4601A

Winter 2026 - Lab #3

Objectives

The goal for this lab is to implement a basic web crawler, crawl some pages, and build a graph representation of those pages. You will also create a basic RESTful server to allow a client to find information about your crawler's output.

As part of the lab, you will have to connect your RESTful server to the lab grading server. Information outlining how to do this is included in the Lab Grader Instructions PDF found under the Labs module in Brightspace. It is recommended you work through these instructions and pass the basic INFO test before completing this lab.

Lab Description

To begin this lab, implement a basic web crawler (e.g., using the [crawler](#) package) that can discover and read pages by branching out from a given seed URL. Your crawler must:

1. Store the content from each page within a database.
2. Store a representation of the network of pages (i.e., which pages link to other pages) within the database. You may store this data in any format you decide is appropriate, such as incoming/outgoing link information stored within each page document, stored as a separate document, stored as a separate collection with link documents, etc.. It is advisable to read the remainder of the requirements below to inform your decisions on data structure and storage.
3. Avoid crawling the same page more than once (i.e., do not re-visit pages).
4. Be capable of crawling multiple datasets and storing separate information for each dataset. Your crawler must crawl the following datasets, starting with the given URL.

Dataset Name	Seed URL
tinyfruits	https://people.scs.carleton.ca/~avamckenney/tinyfruits/N-0.html
fruits100	https://people.scs.carleton.ca/~avamckenney/fruits100/N-0.html
fruitsA	https://people.scs.carleton.ca/~avamckenney/fruitsA/N-0.html

Once your crawler is functioning, implement a basic RESTful server that will allow a client to find information about the data your crawler generates. Your server must support the following requests:

1. GET `/:datasetName/popular` – Returns JSON data containing information for the 10 pages with the highest number of incoming from the given dataset links (i.e., the pages

that receive links from other pages the most). In the case of ties, any of the tied resources can be included in the top 10. The format of the returned JSON data must match the format outlined below. Note that there is a difference between “url”, which refers to the URL on your server the client should request to receive information about a crawled page (see #2), and “origUrl”, which refers to the URL of the webpage that was crawled.

```
{
  "result" : [
    { //result for the most popular page
      "url": "URL to the resource for this page on your server",
      "origUrl": "The original URL that was crawled"
    },
    { //result for the 2nd-most popular page
      "url": "URL to the resource for this page on your server",
      "origUrl": "The original URL that was crawled"
    }
    //...etc...
  ]
}
```

2. If the client makes a GET request for information about a crawled page (i.e., requests one of the “url” entries from #1 above), the server must respond with JSON data with the format below:

```
{
  "webURL": "the original URL that was crawled",
  "incomingLinks": [
    "a list of",
    "incoming links",
    "to the original URL",
    "that was crawled"
  ]
}
```

If you would like to test your crawler, clear your existing database data and start your search from this seed URL: <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-0.html>. This dataset contains 1000 interlinked documents, so if your crawler does not re-visit pages, it should finish running in a relatively short time. The pages with the largest number of incoming links for this dataset are included at the end of the document so you can validate the output of your RESTful server.

While developing/testing, it may be useful to cap the number of pages the crawler visits to a smaller number. An example dataset with only 10 pages can be crawled by starting at the URL <https://people.scs.carleton.ca/~avamckenney/tinyfruits/N-0.html>. For the purposes of this lab, it is acceptable to store all intermediary data (e.g., network structure) in RAM if you wish, but all this data must be saved into the database once the crawl is complete.

Submission

You do not need to submit anything for this lab to Brightspace. To receive grades for this lab, you must connect the RESTful server you have implemented to the central lab grading server and pass the L3 test.

Lab Reflection Questions

You do not need to submit answers to these questions as part of the lab, but they are useful questions to think about.

1. How have you represented the graph of pages in your database?
2. How do you keep track of the incoming/outgoing links during the crawl?
3. Describe the selection policy of your web crawler.
4. Does your crawler store all network information while running and then save it to the database, or does it update the database after each page is visited? Can you describe any advantages/disadvantages of each approach?

Expected Results for fruitgraph Dataset

Top 10 Pages by Incoming Links:

1. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-1.html> (74)
2. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-4.html> (54)
3. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-3.html> (48)
4. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-0.html> (37)
5. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-9.html> (32)
6. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-12.html> (30)
7. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-15.html> (28)
8. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-20.html> (27)
9. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-6.html> (25)
10. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-42.html> (23)

List of Links for Each Top 10 Page:

1. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-1.html> (74):

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-0.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-10.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-102.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-108.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-111.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-12.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-122.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-123.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-126.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-139.html>

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-145.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-155.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-158.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-17.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-183.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-184.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-19.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-196.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-198.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-2.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-228.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-233.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-25.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-27.html>
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<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-34.html>
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<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-5.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-51.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-510.html>
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<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-527.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-538.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-549.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-550.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-574.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-581.html>

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-582.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-614.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-630.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-66.html>
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<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-898.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-91.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-921.html>
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<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-956.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-979.html>

2. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-4.html> (54):

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-1.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-118.html>
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3. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-3.html> (48):

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<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-98.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-991.html>

4. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-0.html> (37):
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-1.html>

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-100.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-104.html>
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<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-854.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-877.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-912.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-937.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-945.html>

5. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-9.html> (32):

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-107.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-11.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-12.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-131.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-147.html>

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-148.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-173.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-22.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-220.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-258.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-26.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-27.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-278.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-295.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-3.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-30.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-38.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-41.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-472.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-5.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-551.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-566.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-582.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-607.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-613.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-64.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-676.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-714.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-797.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-88.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-91.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-950.html>

6. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-12.html> (30):

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-1.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-101.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-113.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-119.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-180.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-246.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-252.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-257.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-259.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-277.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-33.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-333.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-356.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-361.html>

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-379.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-430.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-445.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-448.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-459.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-487.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-545.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-558.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-601.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-615.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-669.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-722.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-741.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-77.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-9.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-95.html>

7. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-15.html> (28):

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-11.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-166.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-187.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-19.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-230.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-247.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-289.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-301.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-326.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-344.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-350.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-351.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-397.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-411.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-445.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-534.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-609.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-645.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-665.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-67.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-674.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-68.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-7.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-729.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-74.html>

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-791.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-924.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-981.html>

8. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-20.html> (27):

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-0.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-142.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-144.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-175.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-192.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-197.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-21.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-28.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-280.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-3.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-416.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-533.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-54.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-569.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-573.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-596.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-62.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-664.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-708.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-742.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-8.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-802.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-808.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-852.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-886.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-936.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-999.html>

9. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-6.html> (25):

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-10.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-151.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-163.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-172.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-229.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-255.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-292.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-303.html>

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-377.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-4.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-526.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-533.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-56.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-560.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-590.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-61.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-63.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-641.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-657.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-67.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-699.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-7.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-888.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-906.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-93.html>

10. <https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-42.html> (23):

<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-0.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-118.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-14.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-146.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-152.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-171.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-173.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-175.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-180.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-201.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-21.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-24.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-243.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-274.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-282.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-318.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-322.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-347.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-441.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-476.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-58.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-687.html>
<https://people.scs.carleton.ca/~avamckenney/fruitgraph/N-812.html>